

Capstone-1 Project Proposal

Project Title

AI Workforce Analytics Dashboard

[Analysing the Impact of Artificial Intelligence on Jobs, Skills, and Hiring Trends.]

1. Summary

This project aims to develop an interactive Power BI dashboard that analyses how Artificial Intelligence is impacting jobs. The dashboard will provide insights into AI replacement risk, future job demand, employee performance, salary distribution, job satisfaction, upskilling requirements, and hiring trends across industries and countries. This dashboard will help organisations make informed decisions regarding workforce planning, employee development, and AI adoption strategies.

2. Problem Statement

Background

Nowadays, Artificial Intelligence is playing the main role; organisations are facing challenges in understanding how AI impacts jobs, employee performance, future workforce requirements and hiring strategies. Businesses require data-driven insights to identify high-risk job roles

Objective

To create a comprehensive Power BI dashboard that helps stakeholders monitor workforce trends, evaluate AI-related risks, and understand employee performance and satisfaction across industries.

Scope

The project will analyse:

- AI Replacement Risk by Job Title and Industry
- Future Demand Risk of Job Roles
- Top Job Roles Risks Percentage
- Country-wise risk
- Salary Analysis across Countries and Industries
- Employee Performance Trends
- Job Satisfaction Analysis

3. Data Sources

Primary Dataset.

Dataset includes:

○ Employee ID
○ Job Title
○ Industry

○ Country
○ Education Level
○ Years of Experience
○ AI Replacement Risk
○ Future Demand Score
○ Remote Work Status
○ Average Salary
○ Required Skills
○ Automation Level
○ Job Growth Rate
○ Work Hours per Week
○ Company Size
○ AI Tool Usage
○ Performance Score
○ Upskilling Required
○ Job Satisfaction
○ Hiring Trend

Source Citation

Dataset Source: Kaggle Dataset {AI Workforce Analytics Dataset}

The dataset will be used for educational and analytical purposes.

4. Methodology

Data Collection

- Download dataset from Kaggle.
- Import data into Power BI.[text_csv]

Data Cleaning

- Transform data into Power Query
- Handle missing values and duplicates.
- Standardise categorical values.

- After cleaning, click on Close and Apply

Data Transformation

- Create required DAX measures and new columns.
- Generate KPIs for workforce analysis.

Dashboard Development

Design multiple dashboard pages:

1. Executive Overview
2. Workforce Demographics
3. Salary Analysis
4. AI Risk Assessment
5. Future Job Demand Analysis
6. Employee Performance Dashboard
7. Job Satisfaction Dashboard
8. Hiring Trends Dashboard

Visualization Techniques

- KPI Cards
- Bar Charts
- Pie Charts
- Table charts
- Slicers and Filters

5. Expected Outcomes

The dashboard will provide:

- Identification of high-risk job roles affected by AI.
- Insights into future-demand occupations.
- Salary comparison across industries and countries.
- Understanding of employee satisfaction and performance.
- More affected job titles by AI
- Hiring trend evaluation for strategic workforce planning.
- Interactive reporting for faster decision-making.

6. Tools and Technologies

- Power BI

- Power Query
- DAX (Data Analysis Expressions)

7. Risks and Challenges

- Missing or inconsistent data values.
- Data quality issues affecting dashboard accuracy.
- Rapid changes in AI trends that may impact findings.

User adoption challenges for dashboard users.

8. Conclusion

The AI Workforce Analytics Dashboard will provide organisations with actionable insights into workforce trends, AI adoption, employee development, and hiring strategies. By leveraging Power BI's interactive capabilities will gain a comprehensive understanding of how artificial intelligence is affecting the job market and workforce requirements. The project aligns with modern HR analytics and supports data-driven decision-making.